

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

CS1A - Actuarial Statistics

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) Let X be a random variable denoting the number of hours you spent on scrolling random social media reels during the day. It has the following cumulant generating function:

$$C_X(t) = 2t + 3t^2$$

Which one of the following is correct?

- A. $E(X) = 2$ and $\text{Var}(X) = 3$
- B. $E(X) = 2$ and $\text{Var}(X) = 6$
- C. $E(X) = 3$ and $\text{Var}(X) = 3$
- D. $E(X) = 2$ and $\text{Var}(X) = 9$
- E. $E(X) = 3$ and $\text{Var}(X) = 3$

[2 marks]

Q.2) The game of musical chairs is played in the following way:



- The first round of the game starts with $(n - 1)$ chairs and n players
- Music is played and players have to go in circle around the chairs
- Music stops and players are supposed to occupy a chair nearest to them
- One player out of the n players is eliminated in that round
- Next round of the game continues with $(n - 2)$ chairs and $(n - 1)$ players
- Game continues till we get the final winner

What is the probability of an individual winning the game of musical chairs?

- A. $1 / (n - 1)$
- B. $(n - 2) / (n - 1)$

- C. $n / (n - 1)$
- D. $(n - 1) / n$
- E. $1 / n$

[2 marks]

Q.3) A civil services aspirant has developed a multi-variate linear regression model which predicts the UPSC mains score (Y) based on two variables:

- X_1 : hours of study per week
- X_2 : number of mock tests taken

The multiple linear regression model is given by:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2$$

If $\alpha = 150$, $\beta_1 = 3$ and $\beta_2 = 2$, which one of the following statements is factually correct?

- A. UPSC mains score would be exactly twice the number of mock tests taken by the aspirant.
- B. For every extra hour of study per week, UPSC mains score increases by 2 marks, holding number of mock tests taken constant.
- C. For every extra mock test taken, UPSC mains score increases by 2 marks, holding weekly hours of study constant.
- D. For every extra mock test taken, UPSC mains score increases by 2 marks, even when the aspirant alters his weekly hours of study.
- E. An aspirant who has not appeared for any mock test, will score zero marks in UPSC mains examination.

[2 marks]

Q.4) In an online retail platform:

- A = number of orders placed in an hour
- B = number of orders with delivery complaints in that hour

The joint probability distribution of A and B is given by:

$$P(A = a, B = b) = (a+3b)/75,$$

where

$$a = 0, 1, 2, 3, 4 \text{ and } b = 0, 1, 2.$$

Given that at least one delivery complaint occurred ($B \geq 1$), what is the expected number of complaints?

- A. 1.38
- B. 1.46
- C. 1.54
- D. 1.62
- E. 1.70

[2 marks]

Q.5) A bank samples 36 customers and finds mean withdrawal = 2,800 with sample variance = 400,000.

What is the standard error of the sample mean?

- A. 632.54
- B. 105.41
- C. 111.17
- D. 95.02
- E. 120.09

[2 marks]

Q.6) Let the rental rate per square feet in a particular locality be denoted by random variable Y . Y follows a lognormal distribution with a known parameter $\sigma^2 = 0.16$.

$Y \sim \text{Lognormal}(\mu, 0.16)$ i.e. $\ln Y \sim N(\mu, 0.16)$

Let the floor level in the building be denoted by random variable X .

We model the mean of $\ln Y$ in a generalised linear model with a single predictor X using the identity link function:

$$E[\ln Y] = \mu = \beta_0 + \beta_1 * X$$

Akshar has finalised a flat on the second floor of a building and is comfortable with the rental rate. However, in the same building there is a flat on the fourth floor which he likes but may not be able to afford.

What is the percentage increase in expected rental rate per square feet i.e. $E[Y]$, if Akshar decides to go with the flat on the fourth floor instead of the flat on the second floor? Assume $\beta_1 = 0.5$.

- A. 50%
- B. 100%
- C. 135%
- D. 172%
- E. 200%

[2 marks]

Q.7) Let X denote the annual budgetary allowance for a particular department in the organization and Y denote the actual spendings by the department during the year.

Let $U = X - Y$ such that U denotes the amount of savings during the year.

Which of the following expressions correctly represents the moment generating function for the random variable U i.e. $M_U(t)$?

Hint: [By definition, joint MGF of (X, Y) is $M_{X,Y}(s_1, s_2) = E[\exp(s_1X + s_2Y)]$.]

- A. $M_{X,Y}(t, -t)$
- B. $M_{X,Y}(t, t)$

- C. $M_X(t) - M_Y(t)$
- D. $M_X(t) * M_Y(-t)$
- E. $\log_e(M_{X,Y}(t, -t))$

[2 marks]

Q.8) A retinologist knows that 2% of patients undergoing a particular eye screening test have a rare eye condition.

In a given year, she screens 2,000 patients. Using the Central Limit Theorem (CLT), what is the probability that more than 50 patients are found to have this condition?

- A. 0.15
- B. 0.08
- C. 0.05
- D. 0.12
- E. 0.20

[2 marks]

Q.9) In a game of ludo, a player needs to roll '6' on a fair six-sided die to move the token out of the starting square. Each roll is assumed to be independent.

Let X be the number of rolls needed to get the first '6'.

What is the probability that the player rolls the die exactly 3 times to roll the first '6'?

- A. $1/36$
- B. $125/216$
- C. $5/18$
- D. $1/6$
- E. $25/216$

[2 marks]

Q.10) Consider the following pairs relating to canonical link functions of some of the common distributions belonging to the exponential family of distributions:

1. Poisson distribution – log link function
2. Gamma distribution – inverse link function
3. Binomial distribution – identity link function
4. Normal distribution – logit link function

Which of the above pairs are correct?

- A. 1 and 3 only
- B. 1 and 2 only
- C. 1, 2 and 3 only
- D. 1, 3 and 4 only

E. 1, 2, 3 and 4

[2 marks]

Q.11) An insurer provides cyber risk cover to small technology firms. For a particular firm –

- The number of cyber incidents in a year follows a Poisson distribution with mean 2.70.
- Each incident results in a loss amount independent of other incidents, with mean of 7,350 and standard deviation of 5,120.

If the insurer wants to set a premium equal to expected loss plus 30% of the standard deviation, what premium should the insurer charge in respect of the above firm?

- A. 22,154
- B. 24,261
- C. 23,502
- D. 25,103
- E. 26,000

[2 marks]

Q.12) Let X denote the height of fellow actuaries attending the global conference of actuaries in the year 2026.

A random sample of nine actuaries ($X_1, X_2, \dots, X_9$) is drawn from a $N(0, \sigma^2)$ distribution. Let

Let $\bar{X}$ and S^2 denote the sample mean and sample variance respectively.

What is the value of

$P(\bar{X} > S)$?

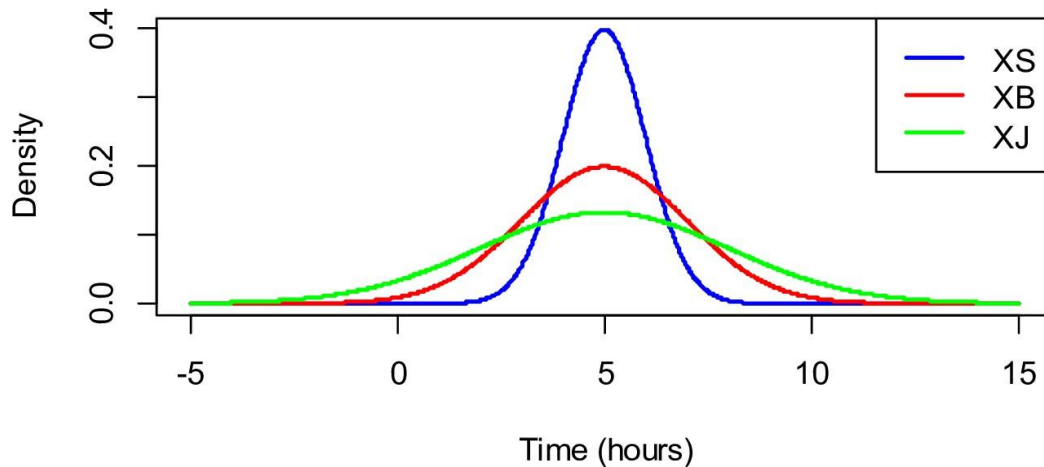
- A. 0.75%
- B. 0.81%
- C. 0.89%
- D. 0.93%
- E. 0.97%

[2 marks]

Q.13) Let X_S, X_B and X_J represent the time taken by ratha yatras of Subhadra-ji, Balaram-ji and Jagannath-ji during the annual ratha yatra festival in Puri. X_S, X_B and X_J are assumed to follow normal distribution with $\mu_S = \mu_B = \mu_J = 5$ hours. However, they do not have equal σ^2 .

The probability density curves for each of the variables are given below:

Normal Distributions (Rath Yatra Time Variability)



As seen in the above graph, X_S has the tallest and narrowest curve whereas X_J has the shortest and widest curve and X_B lies in between.

A new devotee wants to attend the procession this year and wants to know the probability that the ratha yatra time exceeds 6 hours.

Which of the following is TRUE in this context?

- A. $P(X_J > 6) < P(X_B > 6) < P(X_S > 6)$
- B. $P(X_S > 6) > P(X_J > 6) > P(X_B > 6)$
- C. $P(X_J > 6) < P(X_S > 6) < P(X_B > 6)$
- D. $P(X_J > 6) > P(X_B > 6) > P(X_S > 6)$
- E. $P(X_J > 6) = P(X_B > 6) = P(X_S > 6)$

[2 marks]

Q.14) Charlie uses a dating app and each month he has a probability θ of getting a match, independently from month to month.

His prior beliefs about θ follow a Beta (α , β) distribution. Over n months, he observes a total of x matches.

It is known that the posterior distribution of θ is $\theta | x \sim \text{Beta}(\alpha + x, \beta + n - x)$.

If $\alpha = 2$, $\beta = 4$ and $n = 12$, the credibility factor for Charlie's estimate of θ is

- A. 0.6667
- B. 0.8571
- C. 1.0000
- D. 0.2868
- E. 0.8183

[2 marks]

Q.15) In a cosmetics store, the joint probability density function of number of units of skincare cream sold per day (X) and price per unit (P) is given by:

$$f_{X,P}(x,p) = (x+p) / 300, \quad 0 \leq x \leq 10, \quad 0 \leq p \leq 20$$

What is the marginal probability density function of X (no of units sold)?

- A. $(x + 20) / 30$
- B. $(x + 20) / 300$
- C. $(x + 200) / 300$
- D. $(x + 10) / 300$
- E. $(x + 10) / 15$

[2 marks]

Q.16) During an international conflict, a country's border security agency is monitoring the arrival of unmanned reconnaissance drones crossing a sensitive air corridor.

Intelligence analysts are interested in modelling the time between consecutive drone detections at a radar checkpoint.

Let T denote the time (in minutes) between successive drone detections. Based on historical radar data, analysts model this time using an exponential distribution

$$T \sim \text{Exp}(\lambda)$$

where λ is the unknown rate parameter representing the average detection rate of drones

To estimate λ analysts adopt a Bayesian approach. Before observing the new radar data, they assume a Gamma prior distribution for the rate parameter.

$$\lambda \sim \text{Gamma}(\alpha, \beta)$$

where α and β are known parameters.

Suppose that over a monitoring period the radar system records the arrival times between n consecutive drone detections as $t_1, t_2, \dots, t_n$.

i) Write down the likelihood function for the sample $t_1, t_2, \dots, t_n$ given λ . **[2]**

ii) Using the likelihood as determined in part (i) and the Gamma prior, show that the posterior distribution of λ is also a Gamma distribution and verify that its parameters are:

$$\lambda | t \sim \text{Gamma}(\alpha + n, \beta + \sum t_i)$$

[4]

iii) Assume that 20 drone detections have been recorded during the observation period.

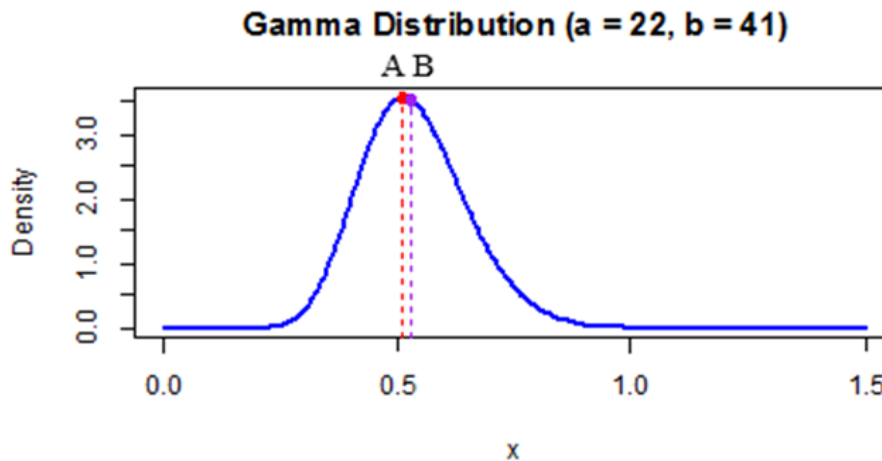
$$\sum_{i=1}^{20} t_i = 40$$

and gamma prior has the parameters $\alpha = 2$ and $\beta = 1$.

Determine the Bayesian estimate for λ under quadratic loss.

[2]

- iv) A plot of the posterior distribution has been given below and two points A (x_A, y_A) and B (x_B, y_B) have been marked on the graph:



At Point A, the value of the gamma density function is the highest, whereas the perpendicular on X axis from Point B divides the area under the gamma density curve into exactly two equal halves.

State, with reasons, whether the following statements are correct or incorrect:

- x_A represents the Bayesian estimate under all or nothing loss;
- x_B represents the Bayesian estimate under absolute error loss.

[2]

[10 marks]

Q.17) An international airport records the number of passengers passing through a particular security checkpoint each hour. Over a monitoring period of 500 hours, the following data were observed:

No of passengers	0	1	2	3	4	≥ 5	Total
No of hours	280	150	50	15	4	1	500

- i) Calculate the sample mean and sample variance for the data. [2]
- ii) Using the method of moments, estimate the parameter of Poisson distribution used to fit to the above data.
Hence, calculate the expected number of hours with 0, 1, 2, 3, 4, and ≥ 5 passengers assuming a Poisson model. [2]
- iii) Based on your answers in parts (i) and (ii), briefly discuss why a Poisson model may not be an appropriate fit for the data. [2]
- iv) Instead of fitting a Poisson distribution to the data, it is decided to use bootstrapping for the purpose of summarising the data.

Following five bootstrap samples viz. B1 to B5 have been generated for this purpose.

No of passengers		0	1	2	3	4	≥ 5	Total
	Sample B1	278	152	49	14	5	2	500
	Sample B2	285	145	52	13	4	1	500
No of hours	Sample B3	276	154	51	15	3	1	500
	Sample B4	282	148	50	16	3	1	500
	Sample B5	279	151	48	15	6	1	500

Calculate the expected number of hours with 0, 1, 2, 3, 4, and ≥ 5 passengers using the bootstrap samples generated above. [2]

- v) Determine whether the method used for generating these samples is parametric (using Poisson parameter determined in part (ii)) or non-parametric. Justify your answer. [2]
[10 marks]

Q.18) A wildlife research unit studying orangutan rehabilitation in Arithmetica examines how annual structured training hours (x) affect the number of minor survival incidents (y) observed post-release. Data from 10 orangutans is given below:

Orangutan	A	B	C	D	E	F	G	H	I	J
Training hours (x)	50	60	65	70	75	80	85	90	95	100
Minor incidents (y)	6	5	5	4	4	3	3	2	2	1

$$\sum x = 770, \quad \sum y = 35, \quad \sum x^2 = 61,600, \quad \sum y^2 = 145, \quad \sum xy = 2,470$$

- i) Determine the equation of the simple regression line. [2]
- ii) Perform an ANOVA test to determine whether the slope of the regression line is significantly different from zero. Perform the test at 1% level of significance. [4]
- iii) An actuarial student who has recently joined the research unit explores several transformations of the standard linear regression model to improve model fit. Four such transformations are considered, and the coefficient of determination R^2 is computed for each.

Model	Regression Equation	R^2
Log-Log	$\ln(Y) = \alpha + \beta * \ln(X)$	94.40%
Reciprocal	$Y = \alpha + \beta * 1/X$	90.80%
Square Root	$Y = \alpha + \beta * \sqrt{X}$	92.30%
Centred	$Y = \alpha + \beta * (X - \bar{X})$	91.50%

Calculate R^2 for the standard linear regression model and comment on which model out of the five models (one standard model and four transformations of the standard model) is the best fit to the orangutan rehabilitation data. [2]

- iv) Using the linear regression model obtained in part (i), determine the value of the residual term for orangutan E. [2]
[10 marks]

- Q.19)** At an amusement park in Mathematica, engineers are testing the reliability of a giantwheel's maximum height in metres. Due to variations in wind resistance, passenger distribution and mechanical performance, the height fluctuates slightly between the rides.



The radius of the giant wheel is 20 metres and the perpendicular distance between the centre and ground level is 22 metres. Based on a sample collected over the past 300 rides on the giant wheel, the mean distance between the centre and the maximum height reached by the giant wheel is 19 metres and it has a standard deviation of 3 metres.

- i) Calculate a 95% two-sided confidence interval for the maximum height achieved by the giant wheel. (Assume critical value of t-distribution to be 1.96). [2]
- ii) Calculate a 95% two-sided prediction interval for the maximum height to be achieved in the next ride of the giant wheel. Comment on why the 95% prediction interval is wider than the 95% confidence interval. (Assume critical value of t-distribution to be 1.96) [3]
- iii) The Government of Mathematica has come up with a new guideline relating to places of public entertainment which makes it mandatory for giant wheels to have a distance of at least 4 metres from the ground level and caps the maximum height that the giant wheel can achieve to 36 metres.

The giant wheel is restructured in order to comply with the regulations and based on 300 rides on the restructured giant wheel, the mean maximum height reached is 35 metres with a standard deviation of 2 metres.

Assuming normality, calculate the probability that a ride on the restructured wheel exceeds 36 metres, and comment on whether the restructured wheel complies with the new guideline at the 5% level of significance. [2]

- iv) Calculate a 95% two-sided confidence interval for the difference in the mean maximum heights of the original and the restructured giant wheel. (Assume critical value of t-distribution to be 1.96) [3]
[10 marks]

Q.20) A health actuary wants to model mental health score (MHS) which is measured on a continuous scale of 0 to 100. He has chosen to use generalised linear models for this purpose and has identified the following five predictors to be used for calculating MHS:

1. REM: Extent of REM sleep (hours / night)
2. DPM: Dopamine levels in blood (ng/mL)
3. MDT: Number of hours spent in meditation / mindfulness per day
4. MRT: Marital status (1 = single, 2 = married, 3 = divorced, 4 = widowed)
5. EMP: Employment status (1 = student, 2 = employed, 3 = unemployed, 4 = retired, 5 = self-employed)

i) Which of the above predictors are factor variables? Justify your answer. [1]

ii) The actuary is trying to use five different models for estimating the mental health score (MHS):

- Model 1: REM + DPM
- Model 2: REM + DPM + MDT + MRT
- Model 3: REM + DPM + MDT + MRT + MDT:MRT
- Model 4: REM + DPM + MDT + MRT + EMP + MDT:MRT + MRT:EMP + MDT:EMP
- Model 5: MRT*EMP*REM

where –

- predictor1: predictor2 means the interaction term between the two predictors
- predictor1*predictor2 includes the main effect as well as the interaction terms between the two predictors
- base scenario reflects marital status = 1 and employment status = 1
- intercept parameter captures the main effects relating to base scenario

Determine the number of parameters in respect of each of the above models. [7]

iii) Based on data collected for a cohort of policyholders, following values are generated for each of the above models including a saturated model (denoted as Model 0) as presented in the below table:

Model	Scaled Deviance	Difference in Scaled Deviance	Critical Point at 5% level of significance
0	99		
1	91	8	5.99
2	80	11	9.49
3	71	9	7.82

4	51	20	31.41
5	34	17	18.31

Identify the model which is the best fit to the data at the 5% level of significance. [2]
[10 marks]

Q.21) A non-life insurer introduces a fraud-detection system to identify suspicious motor insurance claims. The system flags a claim as “fraudulent” if it exceeds certain risk thresholds. In reality, only 8% of claims are fraudulent.

During the observation period, a total of 10,000 claims (fraudulent and genuine combined) are reported.

Following matrix has been developed in respect of the same:

	Predicted Fraud	Predicted Genuine	Total
Actual Fraud	True Positives	False Negatives	Actual Positives
Actual Genuine	False Positives	True Negatives	Actual Negatives
Total	Predicted Positives	Predicted Negatives	Total

The insurer has identified that –

- Sensitivity = 85% (correctly flags fraud when present) and
- Specificity = 90% (correctly clears genuine claims)

i) Explain **Type I error** and **Type II error** in the context of this fraud-detection system. What do these errors mean for the insurer’s operations? [1]

ii) Calculate –

- Expected number of fraudulent claims correctly flagged (true positives).
- Expected number of genuine claims wrongly flagged (false positives)

[1]

iii) The insurer wants to formally test whether the fraud-detection system’s sensitivity is significantly greater than 80% and is working on the following hypotheses:

$$H_0: p = 0.80 \text{ v. } H_1: p > 0.80$$

where p is the true sensitivity

In the last year, out of a total of 800 actually fraudulent claims, if 680 were correctly flagged, perform the hypothesis test at 1% level of significance.

Hint: [Use chi-square test.] [4]

iv) Construct a 95% two-sided confidence interval for the fraud detection system’s sensitivity. Also comment briefly on whether this confidence interval supports the claim that sensitivity exceeds 80%. [4]

[10 marks]

Q.22) A space agency has collected data from 10 spacecraft missions to Jupiter to study the factors affecting mission performance. The dataset includes the following variables:

Mission	Launch mass (tonnes)	Fuel Efficiency (km / kg)	Travel Time (no. of months)	Number of instruments	Mission Cost (millions CU)
J1	2.79	1.46	33.30	13	714.20
J2	5.05	0.95	28.60	18	1121.80
J3	3.34	1.18	27.40	7	1120.60
J4	5.47	1.07	35.90	8	1053.30
J5	5.73	0.60	27.70	18	728.70
J6	1.71	1.40	29.00	5	708.20
J7	3.88	0.75	25.10	15	849.60
J8	5.52	0.54	26.30	11	1198.90
J9	3.98	0.83	18.90	9	899.00
J10	3.55	1.45	15.50	16	1786.70

i) Identify the right option which correctly describes the nature of the dataset. Justify your answer:

- A. Longitudinal data;
- B. Censored data;
- C. Cross-sectional data;
- D. Truncated data.

[2]

ii) Calculate the Karl Pearson's correlation coefficient between launch mass (x) and travel time (y). Also, interpret the results in terms of mission planning.

$$\sum x = 41.02, \quad \sum y = 267.70, \quad \sum x^2 = 184.0878, \quad \sum y^2 = 7493.87, \quad \sum xy = 1106.348$$

[2]

iii) It is decided to calculate the Kendall's correlation coefficient between number of instruments and mission cost. In this context, it is observed that –

- the number of instances “where higher number of instruments lead to higher mission costs”
is lower than
- the number of instances “where higher number of instruments leading to lower mission costs”

Without performing any additional calculations, answer the following questions:

- a) What do you mean by concordant and discordant pairs in this context?

- b) Will Kendal's correlation coefficient between number of instruments and mission cost be positive or negative in this case? Justify your answer.
- c) In the next five years, another ten space missions are expected to be launched to Jupiter. However, the space missions are expected to be designed with higher number of state-of-art high-cost instruments. Will this improve the Kendal's correlation coefficient? [3]
- iv) Without performing any calculations, explain how the presence of an outlier might affect the slope of a simple linear regression line fitted between launch mass and travel time. [3]
- [10 marks]
